Hardware Flow

Smart Cold-Chain System — Farm-to-Market Closed Cold-Chain | Physical Architecture & Component Flow

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1. Overview

The hardware architecture spans three physical zones: the Central Cold-Storage Facility (grid-powered refrigeration + solar-assisted crate charging), the Intelligent Transport Crate (a self-contained sensing, control and actuation unit), and the Vehicle/Field layer (mobile connectivity and location tracking). The same physical crate moves across all three zones without the produce ever being transferred to a different container.

2. Hardware Architecture Layers

Layer	Function	Key Components
Sensing	Measure environmental conditions	SHT31 (temperature + humidity), warehouse ambient sensors
Control	Process sensor data, decide actions	ESP32-C3 microcontroller
Actuation	Maintain required temperature	TEC / Peltier thermal-maintenance unit
Power	Supply and store energy	LiFePO4 battery + BMS, solar panels, grid refrigeration, PCM thermal buffer
Communication	Send live data to backend	Wireless/GSM/IoT communication module, GPS
Display & Alert	Local status indication	LCD/OLED screen, LED indicator, buzzer

3. Central Cold-Storage Facility — Hardware

- **Grid-based storage racks:** Physical slots/positions, each mapped to a unique grid ID for crate placement.
- **Central refrigeration unit:** Grid-powered compressor/cooling system maintaining warehouse-wide temperature and humidity.
- **Solar panel array + charge controller:** Dedicated to charging individual crate batteries while crates are stored (not warehouse refrigeration, under normal operation).
- **Hybrid inverter:** Bridges solar and grid supply; used to route solar power to refrigeration only during a grid-outage emergency mode.
- **Warehouse environmental sensors:** Fixed sensors monitoring storage-block temperature, humidity and power status.
- **PCM (Phase-Change Material) thermal buffers:** Passive thermal storage in cold-storage blocks to slow temperature rise during a total power outage.
- **Facility gateway/server:** Local hub aggregating crate and warehouse sensor data before uplink to the cloud backend.

4. Intelligent Transport Crate — Bill of Materials

Component	Part	Role
Thermal Insulation	PU / PIR rigid foam	Minimizes heat ingress; reduces actuator load
Sensor Unit	SHT31	Reads internal temperature and humidity
Controller	ESP32-C3	Runs closed-loop control logic and communication
Actuator	TEC / Peltier module	Corrects temperature deviation on demand
Power	LiFePO4 battery + BMS	Powers all crate electronics; BMS protects/reports SOC
Local Display	LCD / OLED + LED / buzzer	On-crate status and abnormal-condition indication
Tracking	Unique Crate ID tag + comms module	Identity, GPS location, and data uplink

Design note: the storage crate and the transport crate are the **same physical unit** — it is lifted out of its warehouse grid slot and placed directly into the vehicle, eliminating a produce-transfer step and its associated cold-chain break risk.

5. End-to-End Hardware Flow (Farm to Market)

Harvest — produce graded, Crate ID assigned

Collection Point — produce loaded into standardized reusable crate

Central Cold Storage — crate placed in grid slot; refrigeration (grid) cools environment

Battery Charging — solar array charges the crate's LiFePO4 battery in-slot

Pre-Dispatch Check — temperature, battery SOC, sensor/comm health verified

Crate Removed from Grid Slot → Loaded into Vehicle (no produce transfer)

Transport — SHT31 → ESP32 → Peltier closed loop maintains setpoint

Communication Module — streams location, temperature, humidity, SOC to backend

Arrival at Market — GPS + Crate ID confirm delivery

6. In-Transit Closed-Loop Control (Hardware Signal Path)

SHT31 Sensor reads current temperature

ESP32-C3 compares reading to target setpoint (e.g. 4°C)

Within range → Peltier held at minimum / off

Rising above range → ESP32 activates Peltier thermal-maintenance unit

Approaching setpoint → ESP32 reduces actuator duty cycle

Feedback loop repeats continuously

The PU/PIR insulation passively limits heat gain so the Peltier only needs to supply the correction, not continuous full-power cooling — this is the main hardware-level battery-saving design.

7. Power-Source Hardware Behaviour

Scenario	Refrigeration Hardware Path	Crate Battery Charging Path
Grid ON, Solar OFF (normal / cloudy)	Grid → central refrigeration	Grid → dedicated battery charger
Grid OFF, Solar ON	Solar → hybrid inverter → refrigeration (emergency)	Solar → crate battery (continues)
Grid OFF, Solar OFF	No active refrigeration — PCM + insulation hold temperature passively	No charging — dispatch crates with sufficient existing SOC

8. Pre-Dispatch Hardware Checklist

- Produce temperature already at required setpoint (crate does not rapidly cool warm produce).
- Battery SOC sufficient for the estimated journey duration.
- Estimated temperature-maintenance runtime > expected journey time.
- SHT31 sensor, ESP32 controller and communication module all responding.
- Crate physically sealed and insulation intact.
- For hilly/remote routes: lighter vehicle selected, fewer crates per trip, battery runtime re-verified against route travel time.